

Sphere Formula

G: finite abstract simplicial complex.

$$U(x) = \{y \in G, x \subset y\} \quad \text{star}$$

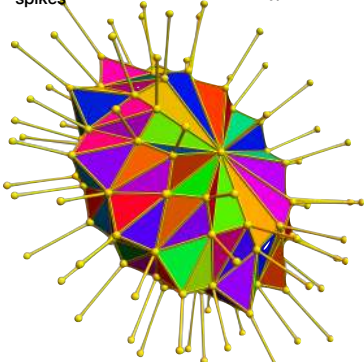
$$S(x) = \delta U(x) = \overline{U(x)} \setminus U(x) \quad \text{unit sphere}$$

$$\sum_{x \in G} \omega(x) \chi(S(x)) = 0$$

Example:

"manifold with spikes"

tips have $\chi(S(x)) = 1$
follicles have $\chi(S(x)) = 1$
hairs have $\chi(S(x)) = 2$



edges within the manifold have $\chi(S(x)) = 0$
triangles within the manifold have $\chi(S(x)) = 0$

Picture

$$S^+(x) = \{y \in G, x \subset y, x \neq y\}$$

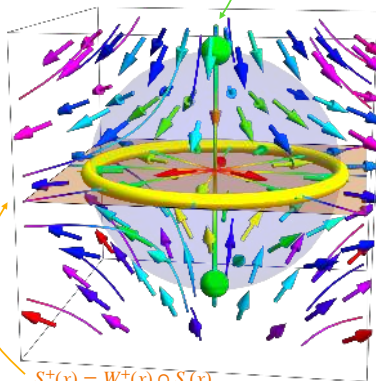
$$S^-(x) = \{y \in G, y \subset x, x \neq y\}$$

$$S(x) = S^-(x) \oplus S^+(x)$$

stable and unstable manifolds intersected with small sphere.

Picture: gradient field of $g(x, y, z) = x^2 + y^2 - z^2$

$$S^-(x) = W^-(x) \cap S_r(x)$$



In the simplicial complex case $g(x) = \dim(x)$

Manifolds

Odd dimensional compact closed manifolds G have $\chi(G) = 0$.



Henri Poincaré
1814-1895

Proof: all $S(x)$ are even dimensional spheres. Euler's Gem shows: $\chi(S(x)) = 2$

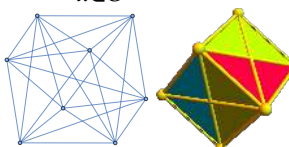
The sphere formula

$$\sum_{x \in G} \omega(x) \chi(S(x)) = 0$$

gives $\sum_{x \in G} \omega(x) = 0$

Example:

$K_{2,2,2,2}$
3-sphere



Hyperbolicity

G: finite abstract simplicial complex.

$$S^+(x) = \{y \in G, x \subset y, x \neq y\}$$

$$S^-(x) = \{y \in G, y \subset x, x \neq y\}$$

stable and unstable spheres.

S^+ open and S^- is closed.

the join of the two complexes is only a delta set but their graph join is $S(x)$.

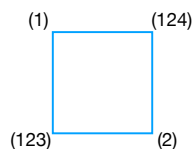
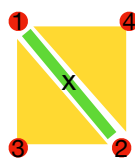
$$S(x) = S^+(x) \oplus S^-(x) \quad \text{join}$$

$x = (12)$

$$U(x) = \{(12), (123), (124)\}$$

$$S^+(x) = \{(123), (124)\}$$

$$S^-(x) = \{(1), (2)\}$$



$$S(x) = \{(1), (2), (123), (124)\}$$

Its graph is C_4

Proof:

G: finite abstract simplicial complex.

$$B(x) = \overline{U(x)} \quad \text{unit ball}$$

$$S(x) = B(x) \setminus U(x) \quad \text{unit sphere}$$

$$B(x) = U(x) \cup S(x)$$

disjoint union of an open and closed set. By counting:

$$\chi(B(x)) = \chi(U(x)) + \chi(S(x))$$

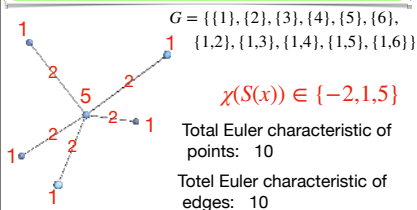
Therefore $\chi(U(x)) = g(x, x)$ the self-energy of a simplex x is

$$\chi(U(x)) = 1 - \chi(S(x))$$

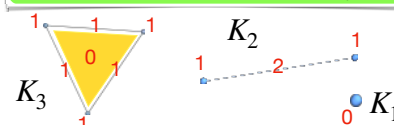
We have seen that $\omega(x)(1 - \chi(S(x)))$ was a curvature adding up to $\chi(G)$

But, $\sum_x \kappa(x) = \chi(G)$ is equivalent to the sphere formula.

Example: Star graph



Example: complete K_n



Example: Poincaré Sphere

The Poincaré sphere is a 3-manifold that has the same cohomology than the 3-sphere but which has a different fundamental group.

$$\chi(S(x)) = 2, \forall x \in G$$

Implementation with $f = (24, 154, 260, 130)$

$b = (1, 0, 0, 1)$

Betti vector

